Benchmark Investigation of Panda: A Pretrained Forecast Model for Chaotic Dynamics

Experiment Log and Analysis

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Anuj Bhadbhade

Indian Institute of Science, Bengaluru

Supervisor: Prof. Flora Salim

University of New South Wales, Sydney

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Abstract

This document records all benchmark experiments conducted on Panda (Lai, Bao, Gilpin, ICLR 2026), a pretrained forecast model for chaotic dynamical systems. The investigation was motivated by Prof. Salim's direction to test Panda's emergent capabilities on standard time series and spatiotemporal benchmarks. Each experiment is reported with its motivation, methodology, methodological limitations, direct observations, and competing explanations. Claims are explicitly labelled as: **[OBS]** observed in data, **[PAT]** empirical pattern across multiple conditions, **[HYP]** proposed explanation, **[SPEC]** speculative, or **[EST]** established by prior theory or paper. This labelling convention was decided to maintain distinction between observation and possible interpretations rather than mixing them.

Contents

1 Experimental Setup

1.1 Models

Two models are compared throughout all experiments.

Panda 21M. The base checkpoint from `GilpinLab/panda` on HuggingFace. A 21 million parameter encoder-only transformer trained exclusively on 20,000 synthetic chaotic ordinary differential equations (ODEs). Prediction horizon 128 steps. Context length 512 steps. For horizons exceeding 128 steps, window autoregression is used: predictions are fed back as context for subsequent rollouts.

Chronos 20M. The `amazon/chronos-t5-small` checkpoint. A 20 million parameter causal univariate decoder model trained on a large corpus of real-world time series. Chronos forecasts each channel independently.

The two models are matched in parameter count, making comparisons attributable to architecture and training distribution rather than scale.

1.2 Evaluation Protocol

Unless stated otherwise, the following protocol is used across all experiments.

Normalisation. Each context window is normalised independently using its own mean and standard deviation per channel (per-window instance normalisation). This prevents global statistics from leaking future information into each evaluation window.

Evaluation windows. A sliding window of context length 512 is moved across the time series. The number of windows is specified per experiment. In early experiments (pre-presentation), $n_{\text{windows}} = 6$–10 was used. In the fixed experiment set, $n_{\text{windows}} = 8$ or $n_{\text{windows}} = 20$ is used as noted.

Primary metric. Mean Absolute Error (MAE) is the primary metric throughout. Symmetric Mean Absolute Percentage Error (sMAPE) was used in earlier experiments but was found to be unreliable on instance-normalised data where values near zero cause the denominator to vanish, artificially inflating or deflating the metric.

Statistical test. Where stated, the Wilcoxon signed-rank test (one-sided, alternative: model A better than model B) is applied to the per-window MAE values. With $n_{\text{windows}} = 8$, the minimum achievable p-value is 0.004. With $n_{\text{windows}} = 20$, the minimum achievable p-value is approximately 0.0001. All p-values should be interpreted in light of these floors.

Advantage. Panda advantage is defined as $\text{MAE}_{\text{Chronos}} - \text{MAE}_{\text{Panda}}$. Positive values indicate Panda is better.

1.3 Hardware

All experiments were run on a CPU-only laptop. Runtimes are consequently long for experiments involving many channels or long horizons.

2 Pre-Presentation Experiments

These experiments were conducted prior to the group meeting and form the basis of the initial findings presented to Prof. Salim.

2.1 Experiment 1: Standard Time Series Benchmarks (Initial)

2.1.1 Motivation

To establish whether Panda has any advantage on non-chaotic standard benchmarks. ETTh1, ETTh2, and Weather were chosen as the canonical time series forecasting benchmarks used by TimesNet, PatchTST, and Chronos comparisons in the literature.

2.1.2 Method

Datasets: ETTh1 (7 channels, 17,420 timesteps), ETTh2 (7 channels, 17,420 timesteps), Weather (21 channels, 52,696 timesteps). Prediction horizon: 128 steps. $n_{\text{windows}} = 10$. Primary metric: sMAPE. Instance normalisation applied globally to the full dataset before windowing (not per-window).

2.1.3 Methodological Limitations

1. sMAPE is unreliable on instance-normalised data where channel values are near zero. Results are not quantitatively trustworthy.
2. Global normalisation leaks future statistics into evaluation windows.
3. Single prediction horizon (128 steps). Results may not generalise to other horizons.
4. No statistical significance tests.
5. $n_{\text{windows}} = 10$ provides limited statistical power.

2.1.4 Observations

[OBS] Both models produced sMAPE approximately 0.8–0.9 on all three datasets. **[OBS]** Panda slightly worse than Chronos on ETTh1. **[OBS]** Chronos sMAPE = 0.009 on Lorenz rho = 15 was flagged as a metric artifact rather than genuine model performance.

2.1.5 Explanation Shipped at the Time

The conclusion presented was: "Panda fails on non-chaotic data because it was trained on stationary chaotic ODEs with no seasonal inductive bias." This was stated as a finding despite the invalid metric and the limitation to a single prediction horizon.

2.1.6 Alternative Explanations Not Presented

1. **[HYP]** The result may be specific to H = 128. Later fixed experiments showed Panda wins on Weather at all tested horizons and on ETTh1 and ETTh2 at longer horizons.
2. **[HYP]** The sMAPE artifact may have masked the true comparison entirely, making the result uninterpretable.
3. **[HYP]** Weather and ETT differ substantially in channel count and coupling structure. Grouping them as "non-chaotic data" ignored this distinction.

Confidence: Very Low. The original conclusion is largely contradicted by the fixed Experiment 8.

2.2 Experiment 2: Double Pendulum Graded Noise

2.2.1 Motivation

To quantify the cost of measurement noise on a chaotic system and test whether Panda's advantage over Chronos is robust to noise. This also served as a proxy for the gap between clean synthetic data and noisy real experimental data.

2.2.2 Method

The double pendulum ODE was simulated with Gaussian observation noise at six levels: $\sigma \in \{0, 0.01, 0.05, 0.10, 0.25, 0.50\}$ times the signal standard deviation. One random seed, one initial condition. $n_{\text{windows}} = 8$. sMAPE and MAE both reported.

2.2.3 Methodological Limitations

1. Single seed and single initial condition. One realisation of the trajectory.
2. The real experimental dataset (Asseman et al. 2018) was not obtained. The experiment was described as "synthetic versus real" but was in practice "synthetic versus synthetic with Gaussian noise." Gaussian additive noise is not the same type of nonstationarity as real experimental measurement error.
3. Noise levels were not tested beyond $\sigma = 0.50$.
4. No statistical significance tests.

2.2.4 Observations

[OBS] Panda MAE at $\sigma = 0$: 0.62. Panda MAE stays approximately flat (0.63–0.67) across all tested noise levels up to $\sigma = 0.50$. **[OBS]** Chronos MAE at $\sigma = 0$: 0.71. Chronos MAE peaks at approximately 0.77 at high noise. **[OBS]** Panda consistently lower MAE than Chronos at every tested noise level. **[OBS]** sMAPE panel showed erratic non-monotone behaviour, correctly identified as a metric artifact on normalised data.

2.2.5 Explanation Shipped at the Time

"The random Fourier and polynomial features are smooth functions of the input, making the embedding noise-robust. Takens' theorem is noise-tolerant." This was stated as a conclusion rather than a hypothesis.

2.2.6 Alternative Explanations Not Presented

1. **[HYP]** Chronos uses a discrete tokenisation scheme. Continuous Gaussian noise may specifically disrupt this tokenisation, independently of any property of Panda's architecture.
2. **[HYP]** Panda's predictions may have lower variance by default. A flat MAE noise curve could reflect low-variance predictions rather than genuine noise robustness.
3. **[HYP]** The tested noise levels may be below the breakdown threshold for both models. The claim cannot be extrapolated to higher noise regimes.

Confidence: Medium. for the observation. **Confidence: Low.** for the proposed mechanism.

2.3 Experiment 3: Lorenz Rho Sweep

2.3.1 Motivation

To test whether Panda's advantage specifically tracks the onset of chaos. The Lorenz parameter ρ controls the transition from periodic ($\rho < 24.74$) to chaotic ($\rho \geq 24.74$) dynamics. This provides a controlled test of the chaos-threshold hypothesis.

2.3.2 Method

$\rho \in \{10, 15, 20, 24, 24.74, 26, 28, 35, 45, 60\}$. One trajectory per ρ value, one initial condition. $n_{\text{windows}} = 8$. MAE as primary metric. Maximum Lyapunov exponent estimated using the Rosenstein method. The estimator required three implementation attempts before producing physically reasonable output.

2.3.3 Methodological Limitations

1. The Lyapunov estimator systematically overestimates. For the Lorenz system at $\rho = 28$, the literature value is $\lambda_1 \approx 0.9$ but the estimator returned 1.66. The x-axis of Lyapunov-based figures is not quantitatively reliable.
2. Single trajectory and single seed per ρ value.
3. $n_{\text{windows}} = 8$ with no confidence intervals on advantage values.
4. Results at rho = 10, 15 showed Chronos sMAPE near zero in the periodic regime, correctly identified as an artifact.

2.3.4 Observations

ρ	Regime	$\hat{\lambda}_1$	Panda MAE	Chronos MAE	Advantage
10.00	Periodic	−0.29	≈0.02	≈0.02	≈0
15.00	Periodic	−0.33	≈0.04	≈0.04	≈0
20.00	Near transition	0.20	≈0.09	≈0.31	+0.22
24.00	Near transition	0.33	≈0.10	≈0.44	+0.34
24.74	Onset of chaos	1.75	≈0.18	≈0.65	+0.47
28.00	Chaotic	1.66	≈0.25	≈0.89	+0.64
60.00	Chaotic	1.63	≈0.48	≈1.17	+0.69

Table 1: Lorenz ρ sweep results (approximate values from $n_{\text{windows}} = 8$, one seed).

[OBS] Estimated λ_1 is negative at $\rho = 10, 15$ and jumps to 1.75 at $\rho = 24.74$, consistent with the theoretical bifurcation point. **[OBS]** Panda MAE advantage is approximately zero at $\rho = 10, 15, 20$. The advantage jumps near $\rho = 24.74$ and remains positive across all tested chaotic ρ values. **[OBS]** No tested chaotic ρ value produced a result where Chronos had lower MAE than Panda. **[PAT]** The pattern of advantage increasing with ρ is consistent across the chaotic range, though not strictly monotone.

2.3.5 Explanation Shipped at the Time

"Panda's advantage is specifically activated by the chaotic transition. The positive Lyapunov exponent is necessary and sufficient for Panda's advantage." This was stated as a conclusion.

2.3.6 Alternative Explanations Not Presented

1. [HYP] Signal statistics (variance, spectral content, amplitude) change discontinuously at the bifurcation. Any model better suited to high-variance aperiodic signals would show this pattern, independently of any chaos-specific mechanism. This alternative is not yet ruled out.
2. [HYP] Chronos tokenisation may fail specifically on aperiodic continuous signals, making Chronos worse in the chaotic regime regardless of Panda's capabilities.
3. [HYP] The advantage may be driven primarily by the Koopman dynamics embedding rather than any chaos-specific learning. This is not testable without an ablation removing the dynamics embedding.

What would falsify the claim: A chaotic system where Panda does not win, or a non-chaotic system where Panda wins substantially. The Weather and Burgers sweep results (Experiments 8 and 10) partially address this.

Confidence: Medium. for the observation. **Confidence: Low.** for the chaos-specific mechanism hypothesis.

2.4 Experiment 4: dysts Systems and Advantage versus Lambda1

2.4.1 Motivation

To test Panda's advantage across a broader set of known chaotic systems from the `dysts` library and to measure whether a Spearman correlation between estimated λ_1 and Panda advantage reaches statistical significance.

2.4.2 Method

Systems tested: Lorenz, Rossler, Chua, Duffing, Halvorsen, SprottB, Thomas, Rucklidge, Dadras, Bouali, DequanLi. $n_{\text{windows}} = 8$ per system. Lyapunov estimates via Rosenstein method. Additional systems were added iteratively until the Spearman p-value crossed 0.05.

2.4.3 Methodological Limitations

1. **P-hacking.** Systems were added iteratively until $p < 0.05$ was achieved. The reported Spearman $\rho = 0.370$, $p = 0.044$ is not a valid pre-registered result. The true p-value corrected for sequential testing is unknown but higher.
2. Systems in the `dysts` library are not independent (several share structural similarities).
3. Lyapunov estimates are noisy and systematically biased (see Experiment 3).
4. Single seed per system. No cross-system multiple comparison correction.

2.4.4 Observations

[OBS] Panda wins (lower MAE) on every tested `dysts` chaotic system. **[OBS]** Spearman $\rho = 0.370$, $p = 0.044$ after iterative addition of systems. This p-value is not reliable due to p-hacking. **[OBS]** Two visual sub-clusters: Lorenz family above the trend line, double pendulum below the trend line.

2.4.5 Explanation Shipped at the Time

"Panda has learned something universal about chaotic dynamics. Chaos is necessary and sufficient. In-distribution systems show larger advantage." Overstated given the methodological problems.

2.4.6 Alternative Explanations Not Presented

1. **[HYP]** The correlation may be driven entirely by the Lorenz family cluster. Removing those points may eliminate significance entirely.
2. **[HYP]** The sub-cluster structure (Lorenz family above trend, double pendulum below) suggests training-distribution effects rather than universality. This directly challenges the "universal chaos learning" claim.

Confidence: Low. due to p-hacking. The observation that Panda wins on every tested chaotic system is more robust (**Confidence: Medium.**) but still limited to $n = 8$ windows per system.

2.5 Experiment 5: Burgers Viscosity Sweep (First Version)

2.5.1 Motivation

To test whether the chaos-threshold pattern found in the Lorenz rho sweep extends to partial differential equations (PDEs). Burgers equation viscosity ν controls the transition from smooth diffusion-dominated to chaotic shock-dominated dynamics, analogous to ρ in Lorenz.

2.5.2 Method

$\nu \in \{0.1, 0.05, 0.02, 0.01, 0.005\}$. PCA reduction to 16 channels. $n_{\text{windows}} = 8$. Lyapunov estimated from first PCA component. Solver diverged at $\nu = 1.0$, leaving no non-chaotic baseline.

2.5.3 Methodological Limitations

1. No non-chaotic baseline. Without $\nu \geq 0.5$, it is impossible to verify a threshold effect.
2. Lyapunov estimation returned approximately zero for all conditions. The chaos-threshold connection could not be verified from the data.
3. Non-monotone reversal of advantage at $\nu = 0.02 \to 0.01$ was not discussed.
4. PCA representation changes character as ν changes, confounding the comparison across conditions.

2.5.4 Observations

[OBS] Panda advantage positive at all tested ν values. [OBS] Advantage roughly increases with decreasing ν but with a non-monotone reversal at $\nu = 0.02 \to 0.01$.

Current status: Superseded by the fixed Experiment 10, which includes a non-chaotic baseline and a stable solver.

2.6 Experiment 6: PCA versus Spatial Subsampling (First Version)

2.6.1 Motivation

To compare two methods for converting a 1D spatiotemporal PDE field into channels for Panda: global PCA modes versus raw spatial location subsampling.

2.6.2 Method

Single ν value (0.005). Single spatial location ($x = 0$) for subsampling. $n_{\text{windows}} = 6$.

2.6.3 Critical Flaw

The spatial location $x = 0$ was a near-nodal point of the Burgers field with near-constant dynamics. Both models trivially predicted near-zero values and scored well. No general conclusion about spatial subsampling is possible from one degenerate location. This experiment was rerun in fixed Experiment 12 with four methods including a variance-stratified control.

2.6.4 Observations

[OBS] Both models performed approximately 8× better in absolute MAE on spatial subsampling than on PCA. [OBS] Panda relative advantage larger with PCA (0.093) than subsampling (0.044).

Confidence: Very Low. for any conclusion. Single degenerate location.

3 Fixed Experiments

These experiments address the methodological problems identified in the pre-presentation set. A new notebook (`fixed_experiments.ipynb`) was created. Key improvements: per-window normalisation, increased n_{windows}, Wilcoxon signed-rank tests, multiple prediction horizons, non-chaotic baselines, and removal of oracle leakage.

3.1 Experiment 7: Standard Horizon Evaluation (Week 1, Invalid)

The Week 1 version of standard horizon evaluation retained global normalisation leakage and used $n_{\text{windows}} = 6$ with sMAPE as primary metric. Results were superseded by Experiment 8 and are not reported here.

3.2 Experiment 8: Standard Horizon Evaluation (Fixed)

3.2.1 Motivation

To establish the true performance gap between Panda and Chronos on standard non-chaotic benchmarks across the full range of prediction horizons used in the literature.

3.2.2 Method

Datasets: ETTh1 (7 channels), ETTh2 (7 channels), Weather (21 channels). Horizons: $H \in \{96, 192, 336, 720\}$. $n_{\text{windows}} = 20$. Per-window normalisation. Wilcoxon signed-rank tests. Weather $H = 720$ was not completed due to hardware time constraints.

3.2.3 Observations

Dataset	H	Panda MAE	IQR	Chronos MAE	IQR	Adv (p)
ETTh1	96	0.7269	0.2203	0.6633	0.2388	−0.064 (0.844)
ETTh1	192	0.8185	0.3046	0.7825	0.2742	−0.036 (0.826)
ETTh1	336	0.8571	0.1971	0.9013	0.1859	+0.044 (0.649)
ETTh1	720	0.9921	0.3226	1.0189	0.3845	+0.027 (0.774)
ETTh2	96	0.8736	0.4790	0.9494	0.5583	+0.076 (0.478)
ETTh2	192	0.9697	0.5380	0.9505	0.5730	−0.019 (0.147)
ETTh2	336	0.9255	0.3860	1.1101	0.3831	+0.185 (0.013)*
ETTh2	720	1.1139	0.2194	1.1027	0.3862	−0.011 (0.392)
Weather	96	0.6378	0.1723	0.8115	0.2036	+0.174 (0.000)*
Weather	192	0.7224	0.2385	0.9582	0.2089	+0.236 (0.001)*
Weather	336	0.8481	0.2898	1.0843	0.3507	+0.236 (0.000)*

Table 2: Fixed Experiment 8. * denotes $p < 0.05$ (Wilcoxon, $n = 20$ windows).

[OBS] On ETTh1, no horizon shows statistically significant difference in either direction. All p-values exceed 0.6. The direction of advantage is inconsistent across horizons. **[OBS]** On ETTh2, one horizon (H = 336) shows significant Panda advantage (p = 0.013). Other horizons are not significant and inconsistent in direction. **[OBS]** On Weather, Panda has statistically significant advantage at H = 96, 192, 336 (all p ≤ 0.001). Advantage magnitude is 0.17–0.24, consistent across horizons. **[PAT]** Weather is the only dataset where Panda shows consistent, statistically significant advantage across multiple horizons.

3.2.4 Competing Explanations for the Weather Result

1. [HYP] Weather has 21 channels versus 7 for ETT. More channels provide more information regardless of coupling. Any multivariate model might benefit.
2. [HYP] Weather has genuine multivariate physical coupling (pressure drives wind drives temperature). Channel attention may capture this.
3. [HYP] Chronos is specifically poorly suited to Weather's complex multi-scale spectral structure (10-minute sampling, 144-step daily cycle).
4. [SPEC] Panda handles quasi-chaotic components of Weather (atmospheric turbulence at synoptic scales) better than Chronos due to its chaotic training distribution.

Confidence: High. for the Weather observation ($n = 20$, $p < 0.001$). **Confidence: Low.** for any proposed mechanism.

3.3 Experiment 9: Univariate Ablation on Weather

3.3.1 Motivation

To directly test whether channel attention drives Panda's advantage on Weather. If channel attention is responsible, removing it (by forecasting each channel independently) should substantially reduce Panda's advantage over Chronos.

3.3.2 Method

Multivariate Panda (all 21 channels processed jointly) was compared against univariate Panda (each channel processed independently, suppressing cross-channel attention) at $H \in \{96, 336\}$. $n_{\text{windows}} = 8$.

3.3.3 Observations

H	panda_uni	panda_multi	Adv (uni better)	p
96	0.5541	0.6113	+0.057	0.074
336	0.8467	0.8762	+0.030	0.371

Table 3: Univariate ablation. Positive advantage means univariate Panda has lower MAE than multivariate Panda.

[OBS] At both horizons, univariate Panda has lower MAE than multivariate Panda. [OBS] Neither result is statistically significant at $\alpha = 0.05$ (p = 0.074, p = 0.371). [OBS] Effect size is small (0.03–0.06 MAE difference).

3.3.4 What This Establishes and Does Not Establish

The data is inconsistent with the hypothesis that channel attention drives Panda's Weather advantage. Channel attention does not help and may marginally hurt on Weather. However the effect is small and not significant at $n = 8$ windows. A definitive conclusion requires more windows.

3.3.5 Competing Explanations

1. [HYP] The temporal architecture (non-causal encoder, Koopman patch embedding, fixed prediction head) is sufficient to beat Chronos. Channel attention adds noise on non-chaotic data. **Confidence: Medium.**: consistent with data.
2. [HYP] $n = 8$ is too small to detect a real but small positive channel attention effect. The true effect may be positive but undetectable. **Confidence: Medium.**: cannot rule out.
3. [SPEC] Channel attention trained on chaotic ODE coupling learns patterns counterproductive on periodic non-chaotic data. **Confidence: Low.**: speculative.

Confidence: Medium. for the observation. The channel attention hypothesis is not supported but not conclusively falsified.

3.4 Experiment 10: Burgers Viscosity Sweep (Fixed)

3.4.1 Motivation

To test the chaos-threshold hypothesis for PDEs with a proper non-chaotic baseline. The stable solver prevents divergence at high ν, enabling comparison across the full smooth-to-chaotic range.

3.4.2 Method

$\nu \in \{2.0, 1.0, 0.5, 0.1, 0.05, 0.02, 0.01, 0.005\}$. Adaptive timestep solver (CFL stability condition satisfied at all ν). PCA to 16 channels. $n_{\text{windows}} = 8$. $H = 128$.

3.4.3 Observations

ν	Physical regime	Panda MAE	Chronos MAE	Adv (p)
2.000	Diffusion dominated	0.0152	0.0191	+0.004 (0.273)
1.000	Diffusion dominated	0.0186	0.0568	+0.038 (0.004)*
0.500	Transition	0.0402	0.1023	+0.062 (0.004)*
0.100	Weakly nonlinear	0.1127	0.2239	+0.111 (0.004)*
0.050	Shock forming	0.1374	0.2865	+0.149 (0.004)*
0.020	Strong shocks	0.1394	0.2593	+0.120 (0.004)*
0.010	Strong shocks	0.1375	0.2325	+0.095 (0.004)*
0.005	Strong shocks	0.1509	0.2732	+0.122 (0.004)*

Table 4: Fixed Burgers viscosity sweep. * denotes $p = 0.004$, the minimum achievable with $n = 8$ windows.

[OBS] At $\nu = 2.0$, Panda advantage is +0.004, p = 0.273. Not significant. This is the strongly diffusion-dominated regime. [OBS] At $\nu = 1.0$, p = 0.004. Significant. Burgers at $\nu = 1.0$ is diffusion-dominated and non-chaotic by standard criteria ($\lambda_1 \leq 0$). [OBS] Panda wins significantly at $\nu = 1.0$ and $\nu = 0.5$, which are not chaotic regimes. [OBS] Advantage is not monotonically increasing with decreasing ν. It peaks near $\nu = 0.05$ (0.149), drops at $\nu = 0.02$ (0.120) and $\nu = 0.01$ (0.095), then rises at $\nu = 0.005$ (0.122).

3.4.4 Critical Finding

[PAT] The chaos-threshold pattern observed in the Lorenz ρ sweep does not replicate cleanly in the Burgers viscosity sweep. Panda wins significantly at non-chaotic viscosity values ($\nu = 1.0$, $\nu = 0.5$) where a chaos-specific mechanism would predict no advantage.

3.4.5 Competing Explanations

1. [HYP] Even at $\nu = 1.0$, the PCA channels of the Burgers field are spatially coupled through the diffusion operator. Panda's channel attention may capture coupling that is present even in non-chaotic smooth flows. **Confidence: Medium.**.
2. [HYP] Chronos is poorly suited to PCA modal time series regardless of chaoticity. The advantage reflects Chronos weakness rather than Panda strength in the non-chaotic regime. **Confidence: Medium.**: consistent with decomposition results.
3. [HYP] The single non-significant result at $\nu = 2.0$ versus the significant result at $\nu = 1.0$ may be noise. The true threshold may lie between 1.0 and 2.0 rather than at the expected chaos onset. **Confidence: Medium.**: cannot rule out with $n = 8$.
4. [SPEC] Burgers PCA modes may have spectral characteristics that look dynamically complex to Panda even at non-chaotic ν, due to the mode-mixing nature of PCA on a spatially heterogeneous field.

What would distinguish these: Estimating λ_1 from the PCA component time series at each ν value using the corrected Rosenstein estimator. If $\lambda_1 < 0$ at $\nu = 1.0$ but Panda still wins significantly, explanations 1, 2, or 3 are supported over a chaos-specific mechanism.

Confidence: Medium. for the observation. **Confidence: Low.** for any specific mechanism.

3.5 Experiment 11: FFT Decomposition Preprocessing (Fixed)

3.5.1 Motivation

To test whether removing periodic and trend components from a time series before presenting it to Panda changes Panda's relative advantage. An earlier version of this experiment (Week 1) contained oracle leakage: the future trend and seasonal components were extracted from the full series and given directly to both models. The fixed version uses only context-window statistics.

3.5.2 Method

FFT decomposition applied to each channel of the context window only. Deterministic component (trend plus seasonal harmonics) projected forward naively: linear trend extrapolation plus repetition of the last full seasonal cycle. Both models forecast the residual; the projected deterministic is added back before evaluation. Comparison against vanilla (no decomposition) run in the same evaluation loop. $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

3.5.3 Note on the Invalid Earlier Version

The oracle-leaking version of this experiment reported a 65% reduction in MAE for both models after decomposition. The corrected version shows a 25% increase in Panda MAE on ETTh1 at H = 336. The two results are in opposite directions. This confirms that the earlier result was entirely an artifact of providing future periodic information to both models.

3.5.4 Observations

Dataset	Cond.	H	Panda	Chronos	Adv (p)
ETTh1	Vanilla	96	0.7205	0.8157	+0.095 (0.422)
ETTh1	Decomp	96	0.8032	0.8293	+0.026 (0.578)
ETTh1	Vanilla	336	0.8407	0.8856	+0.045 (0.273)
ETTh1	Decomp	336	1.0474	1.0174	−0.030 (0.770)
ETTh2	Vanilla	96	0.9329	1.0409	+0.108 (0.230)
ETTh2	Decomp	96	0.8856	0.9640	+0.078 (0.273)
Weather	Vanilla	96	0.6082	0.6880	+0.080 (0.020)*
Weather	Decomp	96	0.6771	0.6871	+0.010 (0.230)
Weather	Vanilla	336	0.8762	0.9987	+0.114 (0.004)*
Weather	Decomp	336	1.0609	1.0760	+0.015 (0.020)*

Table 5: Fixed FFT decomposition experiment. * denotes $p < 0.05$.

[OBS] Decomposition increases absolute MAE for both models on ETTh1 at H = 336 (Panda +25%, Chronos +15%) and on Weather at H = 96 (Panda +11%, Chronos +0.1%). **[OBS]** On Weather, decomposition collapses Panda's advantage. Vanilla H = 96: advantage = +0.080, p = 0.020. Decomp H = 96: advantage = +0.010, p = 0.230. **[OBS]** At H = 336 on Weather, the decomp advantage remains statistically significant (p = 0.020) but the effect size is very small (+0.015 versus vanilla +0.114).

3.5.5 Competing Explanations

1. **[HYP]** Panda's advantage on Weather comes from better handling of the periodic and trend components. On the residual, both models are comparable. **Confidence: Medium.**: consistent with all available data.
2. **[HYP]** The naive FFT projection introduces large errors in the projected deterministic component, degrading both models. The result reflects projection error rather than genuine comparison on the residual. **Confidence: Medium.**: the 25% MAE increase on ETTh1 is suspicious and may reflect projection error.
3. **[SPEC]** The periodic component of Weather contains genuine multivariate coupling. Removing it leaves a weakly coupled residual where channel attention is irrelevant.

Confidence: High. for the observation. **Confidence: Low.** for any mechanism.

3.6 Experiment 12: Subsampling Methods (Fixed)

3.6.1 Motivation

To test whether dynamical diversity subsampling of spatial locations gives Panda a larger advantage than uniform or variance-stratified subsampling. The variance-stratified baseline was introduced to control for the confound in the earlier experiment, where a degenerate near-zero-variance location artificially improved both models.

3.6.2 Method

Four methods: (1) Uniform spacing, (2) Variance-stratified uniform (excludes bottom 10% variance locations before uniform spacing), (3) PCA (global spatial modes), (4) Diversity (farthest-point sampling in a dynamical feature space: standard deviation, mean absolute value, 90th percentile amplitude, spectral entropy). $\nu \in \{0.05, 0.005\}$. $N_{\text{channels}} = 16$. $n_{\text{windows}} = 8$. $H = 128$.

3.6.3 Observations

ν	Method	Panda MAE	Chronos MAE	Adv	p
0.05	Uniform	0.0317	0.0765	+0.045	0.004
0.05	Stratified	0.0296	0.0736	+0.044	0.004
0.05	PCA	0.1374	0.2915	+0.154	0.004
0.05	Diversity	0.0317	0.1132	+0.082	0.004
0.005	Uniform	0.0288	0.1696	+0.141	0.004
0.005	Stratified	0.0301	0.1779	+0.148	0.004
0.005	PCA	0.1509	0.3029	+0.152	0.004
0.005	Diversity	0.0300	0.2475	+0.218	0.004

Table 6: Fixed subsampling comparison. All results p = 0.004 (minimum achievable with $n = 8$ windows).

[OBS] All results reach p = 0.004, the minimum achievable Wilcoxon p-value with $n = 8$ windows. The p-values are therefore uninformative for comparing methods against each other. **[OBS]** Panda absolute MAE across Uniform, Stratified, and Diversity methods is nearly identical at both ν values. At $\nu = 0.005$: Uniform = 0.0288, Stratified = 0.0301, Diversity = 0.0300. **[OBS]** Chronos MAE varies substantially across methods. Diversity locations produce higher Chronos MAE (0.2475) than Uniform locations (0.1696) at $\nu = 0.005$. **[OBS]** Diversity advantage (0.218) exceeds Stratified advantage (0.148) at $\nu = 0.005$. However, the difference is entirely attributable to Chronos MAE (0.2475 vs 1.779). Panda MAE is the same (0.0300 vs 0.0301).

3.6.4 Critical Observation

[PAT] Panda's absolute forecasting performance is invariant to the spatial subsampling method (Uniform, Stratified, Diversity). Diversity subsampling increases Panda's relative advantage over Chronos by making the problem harder for Chronos, not easier for Panda.

3.6.5 Competing Explanations for the Diversity Effect

1. **[HYP]** Diversity locations are more dynamically informative. Chronos cannot handle them but Panda can, demonstrating a genuine advantage of Panda's architecture. **Confidence: Low.**: speculative, not supported by Panda's flat absolute MAE.
2. **[HYP]** Diversity sampling selects spatially extreme or high-variance locations that break Chronos pattern matching without providing additional information to Panda. **Confidence: Medium.**: consistent with the observation that Panda MAE is unchanged.

3. **[HYP]** PCA gives the largest relative advantage because global modes are harder for Chronos (high complexity, global structure) while Panda handles them adequately due to the Koopman embedding. **Confidence: Medium.**.

Confidence: Medium. for the invariance of Panda absolute MAE observation. **Confidence: Low.** for any proposed mechanism.

4 Mechanistic Investigation Experiments

These experiments were designed to address the open scientific questions identified after the fixed experiment set. The primary questions were: (1) what drives Panda's Weather advantage, (2) whether signal statistics confound the Lorenz result, (3) whether $\lambda_1 < 0$ at non-chaotic Burgers viscosities, (4) whether diversity subsampling conclusions are seed-stable, and (5) whether improved period projection changes the decomposition result.

4.1 Experiment 13: Periodic Component Forecasting on Weather (Constructed Target)

4.1.1 Motivation

To test whether Panda's Weather advantage is specifically located in the periodic/deterministic component of the signal. If Panda's advantage amplifies when both models are given only the periodic component as input and evaluated against a periodic target, this suggests Panda handles periodicity better than Chronos.

4.1.2 Method

FFT extraction of the top-5 frequency components (excluding DC) from each context window channel. The resulting periodic context is presented to both models. The target is constructed as the FFT sinusoidal extrapolation of those 5 harmonics into the prediction window — not the real future values of the series. $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

4.1.3 Critical Methodological Note

The target in this experiment is a constructed FFT projection, not the real future values. Both models are therefore evaluated against a deterministic sinusoidal extrapolation rather than ground truth. This conflates "which model better reproduces FFT sinusoidal extrapolation" with "which model better forecasts the actual periodic component of the future." This confound was identified after the run and led directly to Experiment 18 (Option A), which corrects it.

4.1.4 Observations

Condition	H	Panda advantage	p
Full signal (reference)	96	+0.1893	0.004
Periodic only (constructed target)	96	+0.5220	0.004
Full signal (reference)	336	+0.1031	0.020
Periodic only (constructed target)	336	+0.5014	0.004

Table 7: Experiment 13. Periodic-only advantage computed against FFT-constructed target, not real future values. Result superseded by Experiment 18.

[OBS] Panda advantage is approximately 2.8× larger on the periodic-only condition than the full signal at H = 96, and 4.9× larger at H = 336. **[OBS]** Both conditions are statistically significant (p = 0.004 and p = 0.020).

Status: Confounded by the constructed target. Superseded by Experiment 18. The amplified advantage is not interpretable in isolation.

Confidence: Very Low. for any conclusion about Panda's periodic handling ability from this experiment alone.

4.2 Experiment 14: Lorenz Phase Surrogate Control

4.2.1 Motivation

To test whether Panda's advantage on chaotic Lorenz ($\rho = 28$) is specific to chaotic dynamical structure, or whether it is driven by signal statistics alone (variance, power spectrum). A phase-randomization surrogate matches the power spectrum of the original signal exactly while destroying all phase relationships and temporal ordering.

4.2.2 Method

Phase-randomization surrogate applied to the Lorenz $\rho = 28$ trajectory: FFT magnitudes are preserved; phases are replaced with uniform random draws on $[0, 2\pi]$; inverse FFT is applied. The surrogate is rescaled to exactly match the original mean and standard deviation. Both models are run on the original Lorenz and on the surrogate. $n_{\text{windows}} = 8$. $H = 96$.

4.2.3 Methodological Limitations

1. $n_{\text{windows}} = 8$ provides low statistical power. Minimum achievable p-value is 0.004.
2. Phase shuffling introduces endpoint discontinuities and artificial stationarity not present in the original signal. These properties differ from the original for reasons beyond merely removing dynamical structure.
3. The log proposed matching variance and spectral entropy to the $\rho = 15$ periodic baseline. The implemented control is a phase surrogate of $\rho = 28$ itself. These are related but distinct controls.

4.2.4 Observations

Condition	Panda advantage	p
Lorenz $\rho = 28$ (chaotic)	+0.3835	0.004
Phase-shuffled surrogate	+0.1715	0.320

Table 8: Experiment 14. Phase surrogate control for Lorenz $\rho = 28$. $n_{\text{windows}} = 8$.

[OBS] Panda advantage drops by approximately 55% on the surrogate (0.3835 $\rightarrow$ 0.1715).
[OBS] Panda advantage on the surrogate is not statistically significant (p = 0.320).
[OBS] The residual advantage of 0.1715 is not negligible in absolute terms.

4.2.5 Competing Explanations

1. [HYP] Panda's advantage is specific to chaotic dynamical structure. Phase shuffling destroys temporal ordering and the advantage collapses. **Confidence: Medium.**: direction is consistent, but see alternative 2.

2. **[HYP]** The result is a Type II error. With $n = 8$ windows, the test has low power to detect a true advantage of 0.17. The surrogate advantage may be real but undetectable at this sample size. **Confidence: Medium.**: cannot rule out without larger n.
3. **[HYP]** Phase shuffling introduces artificial stationarity that differentially affects both models, independently of any dynamics-specific mechanism. **Confidence: Low.**: known issue with phase surrogates, magnitude unknown.

What would strengthen this: Rerun with $n_{\text{windows}} = 20$. If the surrogate advantage remains near 0.17 and still non-significant, the dynamics-specific hypothesis is strengthened.

Confidence: Medium. for the directional observation. **Confidence: Low.** for the conclusion that dynamics are the specific driver.

4.3 Experiment 15: Burgers λ_1 Estimation from PCA Components

4.3.1 Motivation

Experiment 10 showed Panda wins significantly at $\nu = 1.0$ (diffusion-dominated, non-chaotic by physical criteria). To directly verify whether $\lambda_1 < 0$ at these viscosities, the corrected Rosenstein estimator was applied to the first PCA component time series at each ν value.

4.3.2 Method

Burgers equation simulated at $\nu \in \{2.0, 1.0, 0.5, 0.1, 0.05, 0.01, 0.005\}$ using the stable adaptive-timestep spectral solver. PCA reduction to 16 channels. Rosenstein estimator applied to first PCA component with embedding dimension $m = 3$, delay $\tau = 1$, Theiler exclusion window $w = \max(0.02N, 10)$. $n_{\text{windows}} = 8$. $H = 96$.

4.3.3 Critical Methodological Note on the Estimator

Two independent reasons render the λ_1 estimates unreliable for this system:

1. **Insufficient embedding dimension.** Rosenstein requires $m \geq 2d + 1$ where d is the attractor dimension. For Burgers at low ν, $d \sim \mathcal{O}(1/\nu)$. At $\nu = 0.005$, d may be $\mathcal{O}(100)$. With $m = 3$, nearest-neighbour search is performed in a 3-dimensional projection of a potentially 100-dimensional attractor. All points appear near all others in this projection, producing a flat divergence curve regardless of the true λ_1.
2. **PCA modes are not generic Takens observables.** Takens' theorem guarantees attractor reconstruction only for a generic scalar observable. PCA modes are constructed to maximise explained variance, which means they capture the dominant stable spatial structure while discarding the high-wavenumber chaotic directions. The first PCA component may genuinely not observe the chaotic dynamics even when they are present.

These issues are not fixable by adjusting estimator parameters without changing the observable. All λ_1 values reported here should be treated as unreliable. Physical regime labels derived from the known Burgers bifurcation structure should be used as the chaos proxy instead.

4.3.4 Observations

ν	Physical regime	$\hat{\lambda}_1$ (unreliable)	Panda advantage	p
2.000	Diffusion dominated	NaN	−0.0015	0.191
1.000	Diffusion dominated	−0.011	+0.0045	0.004*
0.500	Transition	−0.007	+0.0156	0.012*
0.100	Weakly nonlinear	−0.004	+0.1010	0.004*
0.050	Shock forming	−0.003	+0.1187	0.004*
0.010	Strong shocks	−0.001	+0.1510	0.004*
0.005	Strong shocks	−0.001	+0.1003	0.004*

Table 9: Experiment 15. $\hat{\lambda}_1$ values are unreliable due to insufficient embedding dimension and non-generic PCA observable. * denotes p = 0.004, the minimum achievable with $n = 8$ windows. $H = 96$ throughout (versus $H = 128$ in Experiment 10).

[OBS] $\hat{\lambda}_1$ is negative or NaN at every tested ν, including the most shock-dominated regimes. Values cluster in the range $[-0.011, -0.001]$ with near-zero absolute magnitude across all ν. This is inconsistent with physical expectations and confirms estimator failure. **[OBS]** Panda advantage increases monotonically with decreasing ν from $\nu = 1.0$ to $\nu = 0.01$ (+0.0045 to +0.1510). A non-monotone reversal occurs at $\nu = 0.005$ ($+0.1003 < +0.1510$). **[OBS]** At $\nu = 2.0$, Panda advantage is −0.0015 (p = 0.191, not significant). Consistent with Experiment 10 at this regime. **[OBS]** Advantage values at $\nu = 1.0$ and $\nu = 0.5$ are substantially smaller here (+0.0045, +0.0156) than in Experiment 10 (+0.038, +0.062). This is attributable to the different prediction horizon (H = 96 here versus H = 128 in Experiment 10), not a contradiction.

4.3.5 Competing Explanations for the Monotonic ν–Advantage Relationship

1. **[HYP]** Lower ν produces more dynamically complex Burgers fields. Panda's Koopman embedding captures this complexity better than Chronos regardless of whether the field is technically chaotic by λ_1 criteria. **Confidence: Medium.**: consistent with data but mechanism not isolated.
2. **[HYP]** Lower ν produces PCA modes with higher frequency content and sharper transitions. Chronos tokenisation degrades on these modes independently of any chaos-specific factor. **Confidence: Medium.**: consistent with Chronos-weakness pattern seen in Experiment 12.
3. **[HYP]** The ν sweep is confounded by changing PCA explained variance ratios. At different ν, the first PCA mode captures a different fraction of total variance, changing the effective forecast difficulty independently of dynamics. **Confidence: Medium.**: PC1 explained variance was 0.819 at $\nu = 2.0$; values at other ν were not recorded.

Confidence: Low. for any conclusion involving $\hat{\lambda}_1$. The estimator is inappropriate for PCA modal time series of high-dimensional PDEs.

4.4 Experiment 16: Multi-Seed Subsampling Variance

4.4.1 Motivation

Experiment 12 reported diversity subsampling results from a single random seed. The critical observation—that Panda absolute MAE is invariant to subsampling method—was based on one realisation. This experiment tests whether that observation is stable across seeds and whether diversity consistently outperforms stratified subsampling.

4.4.2 Method

Diversity subsampling (farthest-point sampling in dynamical feature space) and Variance-Stratified Uniform subsampling run at 10 seeds (0–9) on Burgers $\nu = 0.05$. $N_{\text{channels}} = 16$. $n_{\text{windows}} = 8$. $H = 96$.

4.4.3 Observations

Seed	Diversity adv	Stratified adv	p (div)	p (strat)
0	+0.0677	+0.0424	0.004	0.004
1	+0.0277	+0.0455	0.004	0.004
2	+0.0849	+0.0707	0.004	0.004
3	+0.1018	+0.0362	0.004	0.004
4	+0.0310	+0.1071	0.004	0.004
5	+0.1101	+0.0610	0.004	0.004
6	+0.0716	+0.0432	0.004	0.008
7	+0.1146	+0.0729	0.004	0.004
8	+0.0826	+0.0528	0.004	0.004
9	+0.0797	+0.0324	0.004	0.004
Mean	+0.0772	+0.0564		
Std	±0.0296	±0.0220		

Table 10: Experiment 16. Advantage per seed for Diversity and Stratified Uniform subsampling. Burgers $\nu = 0.05$, $N_{\text{channels}} = 16$, $H = 96$.

[OBS] Panda advantage is positive at every seed and both methods (all $p \leq 0.008$). The fact of Panda winning is robust to seed choice. **[OBS]** The ordering of diversity versus stratified advantage is inconsistent across seeds. At seed 4, stratified advantage (0.1071) exceeds diversity advantage (0.0310). At seed 7, diversity advantage (0.1146) exceeds stratified (0.0729). No consistent dominance. **[OBS]** Diversity advantage standard deviation (0.0296) is 38% of its mean (0.0772). Stratified advantage standard deviation (0.0220) is 39% of its mean (0.0564). Both methods have high relative variance across seeds.

4.4.4 Critical Observation

[PAT] The single-seed finding from Experiment 12 that diversity subsampling gives larger advantage than stratified is not reliable. Across 10 seeds the two methods are not consistently ordered. The seed variance is large relative to the mean advantage for both methods.

4.4.5 Competing Explanations for High Seed Variance

1. [HYP] Different seeds select genuinely different dynamical regimes from the Burgers field, producing legitimately different forecast difficulties. **Confidence: Medium.**: plausible for diversity method, less so for stratified which is deterministic.
2. [HYP] $n_{\text{windows}} = 8$ is insufficient to average out window-to-window variance. Seed variance and window-sampling variance are confounded. **Confidence: Medium.**: cannot be separated at this sample size.
3. [HYP] The diversity metric (dynamical features plus farthest-point sampling) is sensitive to its random initialisation, producing substantially different channel subsets across seeds. **Confidence: Low.**: not directly verified.

Confidence: Medium. for the seed-instability observation. The single-seed Experiment 12 conclusion about diversity subsampling superiority is not supported.

4.5 Experiment 17: Improved Period Projection in Decomposition

4.5.1 Motivation

Experiment 11 used a naive projection method (repeat last full seasonal cycle) for the deterministic component. One competing explanation for the collapse of Panda's advantage after decomposition was that the naive projection introduces large errors, degrading both models. This experiment tests a better projection: average seasonal pattern over all complete periods in the context window, with phase-aligned tiling. No future information is used.

4.5.2 Method

FFT decomposition as in Experiment 11. For the deterministic component projection, instead of repeating the last period, the dominant period is estimated from the FFT peak of the context window and the average over all complete periods is tiled with phase alignment. Evaluated on ETTh1, ETTh2, and Weather at $H \in \{96, 336\}$. $n_{\text{windows}} = 8$.

4.5.3 Observations

[OBS] Improved projection consistently collapses or reverses Panda's advantage relative to vanilla on Weather. At H = 96: vanilla +0.1592 → improved −0.0380. Chronos wins after improved decomposition. [OBS] At Weather H = 336, the improved projection reduces Panda's advantage from +0.1219 to +0.0299 (p = 0.012, significant but small). [OBS] On ETTh1 and ETTh2, improved projection reduces or reverses Panda's advantage at all tested horizons. [OBS] Panda's absolute MAE increases after improved decomposition relative to vanilla on Weather. The improved projection does not reduce overall error; it changes the relative position of the two models.

[PAT] Across Experiments 11, 17, and 18 (Option A), three independent decomposition experiments consistently show that removing or isolating the periodic component collapses or reverses Panda's advantage. This is the most reproducible finding of the mechanistic investigation.

Dataset	Condition	H	Panda advantage	p
Weather	Vanilla	96	+0.1592	0.008
Weather	Improved proj	96	−0.0380	1.000
Weather	Vanilla	336	+0.1219	0.008
Weather	Improved proj	336	+0.0299	0.012*
ETTh1	Vanilla	96	+0.0618	0.039*
ETTh1	Improved proj	96	+0.0551	0.191
ETTh1	Vanilla	336	+0.1005	0.125
ETTh1	Improved proj	336	−0.0382	0.320
ETTh2	Vanilla	96	−0.0015	0.371
ETTh2	Improved proj	96	−0.0546	0.875
ETTh2	Vanilla	336	−0.0671	0.371
ETTh2	Improved proj	336	−0.1426	0.961

Table 11: Experiment 17. Improved period projection versus vanilla. * denotes $p < 0.05$.

4.5.4 Competing Explanations

1. [HYP] Panda's advantage on Weather requires the full mixed signal (trend, seasonal, and residual components jointly). Any decomposition disrupts the joint representation that Panda's Koopman embedding exploits. **Confidence: Medium.**: consistent with all three experiments.
2. [HYP] Both Panda and Chronos are worse after decomposition because the improved projection still introduces errors, and Panda is more sensitive to input quality than Chronos. This makes Panda appear worse relative to Chronos, but the cause is input degradation rather than structural dependence on the full signal. **Confidence: Medium.**: the absolute MAE increase in both models is consistent with this.
3. [HYP] Chronos is specifically better than Panda at forecasting residuals. Once the periodic component is cleanly removed, Chronos recovers its advantage on the remaining stochastic component. **Confidence: Medium.**: consistent with the reversal at Weather H = 96.

Confidence: High. for the observational pattern. **Confidence: Low.** for any specific mechanism.

4.6 Experiment 18: Periodic Context with Real Targets

4.6.1 Motivation

Experiment 13 reported that Panda's advantage amplifies on periodic-only context. However, the target in that experiment was a constructed FFT projection, not the real future values. Experiment 18 corrects this: both models are given the periodic-extracted context but evaluated against the actual future values of the Weather series. This directly determines whether the amplified advantage in Experiment 13 was real or a target-construction artifact.

4.6.2 Method

FFT extraction of top-5 frequency components from each context window channel (same as Experiment 13). Context presented to both models is the periodic component. Target is the real future values from the Weather series. Normalisation is applied using the statistics of the original (not periodic) context window, so the target normalisation is consistent with vanilla evaluation. $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

4.6.3 Observations

Condition	H	Panda MAE	Chronos MAE	Advantage	p
Vanilla (reference)	96	0.6128	0.8021	+0.1893	0.004
Periodic ctx, real tgt	96	1.2007	1.0473	−0.1534	0.992
Vanilla (reference)	336	0.8754	0.9786	+0.1031	0.020
Periodic ctx, real tgt	336	1.1911	1.1816	−0.0095	0.875

Table 12: Experiment 18 (Option A). Both models evaluated against real future Weather values. Periodic context constructed from top-5 FFT harmonics of context window.

[OBS] With periodic context and real targets, Panda loses to Chronos at H = 96 (advantage −0.1534, p = 0.992) and is essentially tied at H = 336 (advantage −0.0095, p = 0.875). **[OBS]** Both models degrade substantially in absolute MAE relative to vanilla. Panda MAE increases from 0.6128 to 1.2007 (+96%). Chronos MAE increases from 0.8021 to 1.0473 (+31%). **[OBS]** Panda degrades approximately 3× more than Chronos in absolute MAE when the context is reduced to its periodic component. **[OBS]** The amplified advantage in Experiment 13 (+0.52) is fully explained by the constructed target and does not reflect a real advantage of Panda on periodic signals.

4.6.4 Critical Finding

[PAT] The advantage Panda shows on Weather requires the full mixed signal. When given only the periodic component as context and evaluated against real targets, Panda loses to Chronos. This is confirmed across three independent experiments: Experiment 11 (decomp collapses advantage), Experiment 17 (improved decomp also collapses advantage), and Experiment 18 (periodic context with real targets reverses advantage).

4.6.5 Implication for TimesNet-Style Pretraining Direction

This result deprioritises TimesNet-style explicit period decomposition as an architectural modification. The evidence shows Panda performs *worse* on periodic signals in isolation, not better. Teaching Panda explicit period decomposition would address a component where Panda already underperforms Chronos, while potentially disrupting the full-signal joint representation that appears to be the source of its advantage.

4.6.6 Competing Explanations for Panda's Larger Degradation

1. **[HYP]** Panda's Koopman patch embedding is degenerate on smooth periodic inputs. Patches of a sinusoid look nearly identical after normalisation; the embedding cannot

distinguish position in the cycle. The forecast head defaults to a near-constant prediction. **Confidence: Medium.**: consistent with the scale of degradation.

2. [HYP] Chronos's pretraining corpus is predominantly periodic and seasonal real-world data. Smooth sinusoidal inputs are near in-distribution for Chronos. **Confidence: Medium.**: consistent with Chronos degrading less.
3. [HYP] Panda is more sensitive to input amplitude calibration. The periodic context has lower energy than the full signal (residual energy removed), and Panda's fixed prediction head scales poorly under this amplitude mismatch. **Confidence: Low.**: speculative, not directly tested.

Confidence: High. for the core observation. **Confidence: Low.** for any specific mechanism explaining Panda's larger degradation.

5 Architectural Component Investigations

These experiments were designed to move from observational findings toward mechanistic diagnosis. Each experiment targets one identifiable architectural component of Panda or one structural property of the evaluation datasets, motivated by the open questions identified at the end of the mechanistic investigation.

5.1 Experiment 19: Complexity Continuum

5.1.1 *Motivation*

The Lorenz ρ sweep (Experiment 3) showed a pattern consistent with a chaos threshold, but confounded by simultaneously changing signal statistics (variance, spectral content, amplitude). This experiment uses five qualitatively distinct dynamical systems spanning the full range from periodic to chaotic, without the confound of continuously varying a single parameter. The goal is to test whether Panda's advantage tracks dynamical complexity or something else.

5.1.2 *Method*

Five systems: Harmonic oscillator (periodic), Van der Pol oscillator (limit cycle), Duffing oscillator (weakly chaotic), Rossler (chaotic), Lorenz (chaotic). $H = 96$. $n_{\text{windows}} = 8$. Permutation entropy at order 3 computed per system as a complexity proxy.

5.1.3 *Observations*

System	Regime	PE	Panda MAE	Chronos MAE	Adv (p)
Harmonic	Periodic	0.438	0.0647	0.4346	+0.370 (0.004)*
Van der Pol	Limit cycle	0.431	0.0319	0.0426	+0.011 (0.027)*
Duffing	Weakly chaotic	0.476	0.5811	0.7948	+0.214 (0.055)
Rossler	Chaotic	0.442	0.0826	0.3868	+0.304 (0.004)*
Lorenz	Chaotic	0.455	0.0555	0.5314	+0.476 (0.004)*

Table 13: Experiment 19. * denotes $p < 0.05$ (Wilcoxon, $n = 8$ windows).

[OBS] Panda has a statistically significant advantage on the Harmonic oscillator (+0.370, p = 0.004). This is the simplest possible periodic system—a pure sinusoid. This directly contradicts the hypothesis that chaos is necessary for Panda's advantage. **[OBS]** Van der Pol (limit cycle) shows the smallest advantage of all tested systems (+0.011, p = 0.027). Effect size is very small. **[OBS]** Duffing (weakly chaotic) advantage +0.214 is not statistically significant (p = 0.055), just above the threshold. **[OBS]** Rossler and Lorenz (both chaotic) show large, statistically significant advantages (+0.304 and +0.476). **[OBS]** Permutation entropy at order 3 ranges from 0.431 to 0.476 across all five systems. It cannot discriminate periodic from chaotic regimes in this dataset: the PE values are nearly identical across qualitatively different dynamics.

5.1.4 *Critical Finding*

[PAT] The advantage pattern is non-monotone with respect to dynamical complexity. The Harmonic oscillator produces the second-largest advantage (+0.370), larger than the weakly chaotic Duffing (+0.214). The Van der Pol limit cycle produces the smallest advantage (+0.011), which is anomalously low relative to both the simpler Harmonic and the more complex chaotic systems. The chaos-specific hypothesis cannot account for either the large Harmonic advantage or the small Van der Pol advantage.

5.1.5 *Competing Explanations*

1. [HYP] Chronos specifically fails on sinusoidal inputs due to its discrete tokenisation scheme. The large Harmonic advantage reflects Chronos weakness rather than Panda strength. The small Van der Pol advantage may similarly reflect Chronos being relatively adequate for limit cycle dynamics. **Confidence: Medium.**: consistent with data; requires Chronos-specific ablation to confirm.
2. [HYP] Panda's Koopman embedding represents harmonic oscillations efficiently because sinusoidal functions are eigenfunctions of the Koopman operator for linear systems. This would make the Harmonic oscillator in-distribution in a functional sense even though it is not a chaotic ODE. **Confidence: Low.**: speculative, not testable without code access to the lifting.
3. [HYP] The Van der Pol limit cycle has dynamical properties that are close to a transition point between two regimes. The small advantage at this transition may reflect genuine model uncertainty rather than a failure of either model. **Confidence: Low.**: speculative.
4. [SPEC] The Harmonic oscillator advantage may be an artifact of the specific initial conditions or trajectory length chosen, rather than a robust property of the system class. **Confidence: Very Low.**: not tested with multiple seeds.

What permutation entropy at order 3 establishes and does not establish: PE at order 3 is computed on very short embedded sequences. With embedding dimension $m = 3$, only $3! = 6$ ordinal patterns are possible. This is insufficient resolution to distinguish dynamics across qualitatively different systems that all have roughly similar power spectra at the timescales being embedded. The metric is not useful as a complexity discriminator at this order.

Confidence: Medium. for the observational pattern. **Confidence: Low.** for any proposed mechanism. The chaos-specific hypothesis is directly contradicted by the Harmonic oscillator result.

5.2 Experiment 20: Chronos Residual Ablation

5.2.1 *Motivation*

Three decomposition experiments (11, 17, 18) showed that removing the periodic component collapses Panda's advantage. A critical ambiguity remained: does Panda's advantage require the full mixed signal because Panda specifically benefits from the joint structure, or because Chronos is specifically hurt by the periodic component? This experiment directly measures how Chronos performs on Weather residuals versus the full signal, without involving Panda at all.

5.2.2 *Method*

Chronos run on: (1) full Weather signal (vanilla), (2) FFT residual of Weather after removing top harmonics (residual). Both evaluated against the real future values. $n_{\text{windows}} = 8$.

$H \in \{96, 336\}$.

5.2.3 Observations

Condition	H	Chronos vanilla	Chronos residual	Δ	p
Weather	96	0.7695	1.0464	+0.277	0.016*
Weather	336	1.0012	1.2461	+0.245	0.008*

Table 14: Experiment 20. Chronos MAE on full signal versus FFT residual. $\Delta > 0$ means residual is harder for Chronos. * denotes $p < 0.05$.

[OBS] Chronos MAE increases significantly on the FFT residual relative to the full signal at both horizons (+0.277 at H = 96, +0.245 at H = 336; both $p < 0.02$). **[OBS]** The magnitude of Chronos degradation after decomposition (+0.277) is comparable to the magnitude of Panda degradation found in earlier decomposition experiments.

5.2.4 Critical Revision to Earlier Interpretation

[PAT] Both Panda and Chronos degrade approximately equally when the periodic component is removed. The earlier interpretation from Experiments 11 and 17 — that Panda's advantage requires the full mixed signal specifically — must be revised. The decomposition results are equally consistent with projection error in the FFT decomposition degrading both models proportionally. The signal-dependence of Panda's advantage is not specifically established by the decomposition experiments; both models depend on the full signal.

5.2.5 Competing Explanations

1. **[HYP]** FFT decomposition introduces substantial projection error in the deterministic component. Both models receive a degraded context and their MAE increases for this reason, not because either model specifically depends on the periodic component. **Confidence: Medium.**: the similar magnitude of degradation in both models supports this.
2. **[HYP]** Both Panda and Chronos leverage the periodic component for forecasting. The Weather residual is genuinely harder for both models because it contains less predictable structure. **Confidence: Medium.**: also consistent with the data; cannot be distinguished from hypothesis 1 without a perfect oracle decomposition.
3. **[SPEC]** The periodic component of Weather carries the dominant predictable signal (daily cycle, annual seasonality). Removing it leaves a near-unpredictable residual. Both models degrade because the task becomes near-impossible, not because of any model-specific failure. **Confidence: Low.**: speculative without measuring residual autocorrelation structure.

Confidence: High. for the observation. **Confidence: Medium.** for the revision to the earlier interpretation. The decomposition story is now less diagnostic than previously believed.

5.3 Experiment 21: Permutation Entropy as Complexity Predictor

5.3.1 Motivation

To test whether a single complexity metric—permutation entropy at order 3—could predict Panda's advantage across all tested systems simultaneously, including Lorenz, Burgers, ETTh1, ETTh2, and Weather. This would provide a single quantitative criterion for predicting when Panda generalises.

5.3.2 Method

PE at order 3 computed per dataset (or per ρ value for Lorenz, per ν value for Burgers). Advantage values taken from previously completed experiments. A visual and Spearman correlation analysis was performed across the combined dataset.

5.3.3 Critical Methodological Note

The PE values separate into two disjoint regimes with no overlap: Burgers PE ranges from 0.024 to 0.132; real datasets (ETTh1, ETTh2, Weather) range from 0.756 to 0.955; Lorenz values cluster at 0.460–0.494. These three groups do not share a common scale. Any correlation computed across all three groups is dominated by between-group separation rather than within-group structure and cannot be interpreted as a genuine predictor.

5.3.4 Observations

[OBS] Burgers PE values (0.024–0.132) are an order of magnitude lower than real dataset PE values (0.756–0.955). The metric operates in fundamentally different regimes for PDE-derived PCA modal time series versus raw sensor data. **[OBS]** Lorenz PE values (0.460–0.494) are insensitive to ρ, ranging from $\rho = 10$ (periodic, PE = 0.494) to $\rho = 60$ (strongly chaotic, PE = 0.473). PE at order 3 cannot discriminate the periodic from chaotic Lorenz regimes. **[OBS]** Within the Lorenz sweep, Panda advantage varies from 0 to 0.69 while PE varies by less than 0.05. There is no within-group correlation. **[OBS]** Experiment 19 independently confirmed that PE at order 3 cannot discriminate Harmonic, Van der Pol, Duffing, Rossler, and Lorenz systems, which have PE values ranging only from 0.431 to 0.476.

5.3.5 Conclusion

[PAT] Permutation entropy at order 3 is not a useful predictor of Panda's advantage. The metric fails on three independent grounds: (1) it cannot discriminate periodic from chaotic Lorenz dynamics, (2) it cannot discriminate across systems in the complexity continuum, and (3) it operates in incomparable numerical regimes across different data types. A higher-order PE (order 5 or 6) on longer time series with more stringent embedding parameters might partially address (1) and (2), but the between-group scale problem (3) would persist without normalisation.

Confidence: High. for the failure of PE order 3 as a discriminator. The finding is consistent and replicated across three data contexts.

5.4 Experiment 22: Node Identity Embeddings (Scalar Offset Proxy)

5.4.1 Motivation

The heterogeneity stratification results (Section 5.4) identified sensor heterogeneity as a factor correlated with reduced Panda advantage. One architectural hypothesis is that Panda lacks sensor identity information: it treats all channels as interchangeable state variables, whereas real weather sensors have distinct physical meanings. This experiment tests a minimal version of sensor identity: adding a learned scalar offset per channel (a per-sensor bias term) and measuring whether this modifies Panda's relative advantage on homogeneous versus heterogeneous subsets.

5.4.2 Method

A scalar offset is fitted per channel using the context window mean of each channel. This is not a full node embedding; it is the minimal possible channel identity signal (a scalar constant per channel). Two Weather subsets were used: (1) `homo_matched`, a homogeneous subset with very low heterogeneity (0.086), and (2) `hetero_controlled`, a heterogeneous subset with high heterogeneity (0.853) and controlled difficulty variance (CV = 0.11). $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

5.4.3 Observations

Subset	H	Panda base	Panda offset	Chronos	Adv base	Adv offset	Δ Panda
homo_matched	96	0.328	0.412	0.714	+0.386	+0.439	−0.085
homo_matched	336	0.839	0.882	1.207	+0.368	+0.144	−0.043
hetero_controlled	96	0.605	0.568	0.847	+0.242	+0.277	+0.037
hetero_controlled	336	0.867	0.870	0.954	+0.088	+0.146	−0.003

Table 15: Experiment 22. Δ Panda = Panda offset − Panda base. Negative means offset hurts Panda. Chronos MAE is unchanged across conditions (Chronos does not use offsets).

[OBS] On the homogeneous subset at H = 96, the scalar offset hurts Panda MAE (+0.085, Panda gets worse). At H = 336, Panda also gets worse (+0.043). **[OBS]** On the heterogeneous subset at H = 96, the scalar offset marginally improves Panda MAE (−0.037, Panda gets better). At H = 336, essentially no change (−0.003). **[OBS]** The direction of the offset effect is opposite on homogeneous versus heterogeneous subsets at H = 96: hurts on homogeneous, helps on heterogeneous. This is directionally consistent with the hypothesis that channel identity information is only beneficial when channels are genuinely distinguishable. **[OBS]** Effect sizes are small throughout. The largest effect (Δ = −0.085 on homo_matched H = 96) is in the wrong direction for the hypothesis; the small positive effect on hetero_controlled H = 96 (Δ = +0.037) is marginal.

5.4.4 Competing Explanations

1. **[HYP]** A scalar offset is too minimal to encode channel identity in any useful sense. The null result reflects the weakness of the proxy rather than the absence of a channel identity effect. **Confidence: Medium.**: consistent with the small effect sizes.

2. [HYP] Channel identity information is not what is missing from Panda on heterogeneous data. The heterogeneity bottleneck identified in Sections 5.4 and 5.5 may be due to interaction structure (graph topology) rather than per-sensor identity. **Confidence: Medium.**: cannot distinguish these from this experiment.
3. [HYP] Adding a per-channel mean offset disrupts the per-window normalisation scheme, which already removes per-channel mean by construction. The offset may be double-correcting and introducing noise on homogeneous channels that have well-defined means. **Confidence: Low.**: requires analysis of the normalisation interaction to confirm.

Confidence: Low. for any conclusion about sensor identity. The scalar offset is too minimal a proxy and the effect sizes are too small to support strong claims.

5.5 Experiment 23: Prediction Head Fine-Tuning

5.5.1 Motivation

Panda's prediction head was pretrained on chaotic ODE trajectories and is fixed at inference. A hypothesis is that the head imposes a chaotic prior on the output distribution, limiting performance on non-chaotic data. This experiment tests whether fine-tuning only the prediction head (leaving the encoder frozen) on Weather improves Panda's performance.

5.5.2 Method

The prediction head alone was fine-tuned on Weather context-forecast pairs for 50 gradient steps. Encoder weights were held fixed. Performance compared against the base Panda (no fine-tuning) at $H \in \{96, 336\}$. $n_{\text{windows}} = 8$.

5.5.3 Observations

H	Panda base	Panda FT	Adv base	Adv FT	Δ Panda
96	0.609	0.699	+0.154	+0.071	−0.091
336	0.870	0.881	+0.189	+0.198	−0.012

Table 16: Experiment 23. Δ Panda = Panda FT − Panda base. Negative means fine-tuning hurts. Adv = $\text{MAE}_{\text{Chronos}} - \text{MAE}_{\text{Panda}}$.

[OBS] Head fine-tuning at H = 96 worsens Panda MAE by 0.091. This is a substantial degradation (approximately 15% relative increase in MAE). [OBS] At H = 336, the degradation is small (−0.012) and likely not significant at $n = 8$ windows. [OBS] Panda's relative advantage over Chronos decreases after fine-tuning at H = 96 (+0.154 → +0.071). [OBS] The direction of the H = 96 result is strongly negative. Fine-tuning the prediction head without fine-tuning the encoder worsens performance.

5.5.4 Critical Finding

[PAT] Prediction head fine-tuning at H = 96 produces a negative result: Panda performance degrades. This rules out the prediction head as an easily recoverable bottleneck through lightweight adaptation.

5.5.5 *Competing Explanations*

1. **[HYP]** The encoder and head co-adapted during pretraining. Fine-tuning the head alone without updating the encoder introduces a mismatch between encoder representations and head expectations. The head adapts to the target distribution but the encoder still produces representations calibrated for chaotic ODE distributions. **Confidence: Medium.**: consistent with the magnitude of degradation.
2. **[HYP]** 50 gradient steps is insufficient for genuine adaptation but sufficient to corrupt the pretrained head. The degradation reflects the head moving away from its pretrained state without reaching a new stable optimum. **Confidence: Medium.**: cannot rule out without testing more gradient steps.
3. **[HYP]** The prediction head is not the bottleneck. The representation upstream of the head (encoder output) contains the relevant distributional mismatch, and adapting the head cannot address this. **Confidence: Medium.**: consistent with the overall pattern across experiments that no single downstream component is responsible.

Confidence: Medium. for the observational result. **Confidence: Low.** for any specific mechanism. The head-fine-tuning direction is not supported as a lightweight adaptation strategy.

6 Sensor Heterogeneity Investigation

These experiments address the hypothesis that Panda's advantage on Weather is partially limited by sensor heterogeneity—the degree to which Weather channels represent qualitatively different physical processes with different dynamical properties—whereas Panda's pretraining distribution consists of homogeneous state variables from single dynamical systems.

6.1 Experiment 24: Heterogeneity Stratification on Weather

6.1.1 *Motivation*

To test whether Panda's advantage over Chronos systematically decreases as the dynamical heterogeneity of the channel subset increases. Weather has 21 channels spanning temperature, humidity, pressure, wind speed, and other variables with distinct dynamical properties. Subsets of Weather channels can be constructed with controlled levels of heterogeneity.

6.1.2 *Method*

Three subsets of 7 Weather channels each were constructed by clustering channels on dynamical feature vectors (standard deviation, lag-1 autocorrelation, spectral entropy, dominant frequency, percentile amplitude): (1) `homogeneous` (all channels from the same cluster, heterogeneity index 0.035), (2) `mixed` (channels from two clusters, heterogeneity index 0.668), (3) `heterogeneous` (channels from maximally distinct clusters, heterogeneity index 0.947). $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

6.1.3 *Observations*

Subset	Het.	H	Panda MAE	Chronos MAE	Adv	p
homogeneous	0.035	96	0.318	0.689	+0.371	0.004*
homogeneous	0.035	336	0.817	1.183	+0.365	0.004*
mixed	0.668	96	0.581	0.786	+0.205	0.039*
mixed	0.668	336	0.925	0.982	+0.056	0.371
heterogeneous	0.947	96	0.618	0.697	+0.079	0.074
heterogeneous	0.947	336	1.123	1.163	+0.040	0.191

Table 17: Experiment 24. Het. = heterogeneity index. * denotes $p < 0.05$.

[OBS] Panda advantage on the homogeneous subset at H = 96 is +0.371 (p = 0.004). On the heterogeneous subset at H = 96, advantage is +0.079 (p = 0.074, not significant). Advantage drops by approximately 5× from homogeneous to heterogeneous at the same horizon. **[OBS]** Panda absolute MAE increases monotonically with heterogeneity at H = 96 (0.318, 0.581, 0.618). Chronos MAE also increases (0.689, 0.786, 0.697) but non-monotonically. **[OBS]** At H = 336, the homogeneous subset shows significant advantage (+0.365, p = 0.004). Mixed and heterogeneous subsets are not significant (p = 0.371, p = 0.191). **[OBS]** The confound is that homogeneous, mixed, and heterogeneous subsets may also differ in difficulty: the homogeneous subset may consist of easier-to-forecast channels.

6.2 Experiment 25: Difficulty-Matched Heterogeneity Control

6.2.1 *Motivation*

The stratification result in Experiment 24 is confounded by channel difficulty. The homogeneous subset channels may be intrinsically easier to forecast, regardless of heterogeneity. This experiment creates a difficulty-matched version of the comparison by ensuring all three subsets have approximately equal individual-channel Chronos MAE.

6.2.2 *Method*

Difficulty was estimated per channel as Chronos univariate MAE. Channels were selected for each heterogeneity level such that the mean difficulty was matched across subsets. Three matched subsets: `homo_matched` (heterogeneity 0.086, mean difficulty 0.800), `mixed_matched` (heterogeneity 0.602, mean difficulty 0.839), `heterogeneous` (heterogeneity 0.947, mean difficulty 0.839). $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

6.2.3 *Observations*

Subset	Het.	H	Panda MAE	Chronos MAE	Adv	p
homo_matched	0.086	96	0.331	0.700	+0.369	0.004*
homo_matched	0.086	336	0.841	1.147	+0.305	0.004*
mixed_matched	0.602	96	0.475	0.808	+0.333	0.004*
mixed_matched	0.602	336	0.856	0.946	+0.090	0.020*
heterogeneous	0.947	96	0.618	0.836	+0.218	0.074
heterogeneous	0.947	336	1.123	1.243	+0.120	0.125

Table 18: Experiment 25. Difficulty-matched heterogeneity control. * denotes $p < 0.05$.

[OBS] After difficulty matching, the pattern from Experiment 24 persists. Panda advantage on homo_matched at H = 96 is +0.369 (p = 0.004). Heterogeneous advantage at H = 96 is +0.218 (p = 0.074, not significant). Advantage drops by approximately 1.7×. **[OBS]** Panda absolute MAE still increases monotonically with heterogeneity (0.331, 0.475, 0.618 at H = 96) despite matched Chronos difficulty. Chronos MAE is approximately constant across subsets (0.700, 0.808, 0.836) after difficulty matching. **[OBS]** The heterogeneity effect on Panda MAE remains after controlling for individual channel difficulty. Chronos MAE is relatively stable; Panda MAE increases strongly with heterogeneity.

6.2.4 *Critical Finding*

[PAT] After controlling for channel-level forecasting difficulty, Panda MAE increases monotonically with sensor heterogeneity while Chronos MAE remains approximately stable. This pattern cannot be explained by difficulty alone. Sensor heterogeneity has a specific negative effect on Panda that is not shared by Chronos.

6.2.5 *Competing Explanations*

1. **[HYP]** Panda treats channels as interchangeable state variables. On homogeneous subsets, this is appropriate. On heterogeneous subsets, cross-channel attention

produces incorrect coupling estimates between physically unrelated sensors. **Confidence: Medium.**: directionally consistent with difficulty-matched result and with the univariate ablation result (Experiment 9) showing univariate Panda competitive with multivariate.

2. [HYP] Difficulty matching by individual Chronos MAE does not fully control for interaction complexity. Heterogeneous subsets may have higher interaction complexity, requiring more sophisticated cross-channel reasoning that neither model handles well but that specifically breaks Panda's joint attention. **Confidence: Medium.**: cannot distinguish from hypothesis 1 with current data.
3. [HYP] Heterogeneous subsets have higher variance in their dynamical feature distributions. Per-window normalisation removes per-channel mean and variance but not higher-order distributional differences. Panda's fixed embedding may be more sensitive to these residual differences than Chronos. **Confidence: Low.**: speculative.

Confidence: Medium. for the core observational finding. The difficulty-matched control substantially strengthens the heterogeneity pattern but does not establish a mechanism.

6.3 Experiment 26: Variance-CV Heterogeneity Control

6.3.1 Motivation

A remaining confound is that difficulty-matched heterogeneous subsets may still contain channels with higher within-subset variance of difficulty (more difficult channels spread more unevenly). This experiment adds a second-order difficulty control by matching the coefficient of variation (CV) of difficulty across subsets.

6.3.2 Method

The heterogeneous controlled subset (`hetero_controlled`) was constructed to have both high heterogeneity (0.853) and low difficulty CV (0.110), matching the difficulty CV of the homogeneous subsets. $n_{\text{windows}} = 8$. $H \in \{96, 336\}$.

6.3.3 Observations

Subset	Het.	H	Panda MAE	Chronos MAE	Adv	p
hetero_controlled	0.853	96	0.605	0.866	+0.261	0.004*
hetero_controlled	0.853	336	0.867	1.175	+0.309	0.012*

Table 19: Experiment 26. Heterogeneous subset with controlled difficulty variance (CV = 0.11). * denotes $p < 0.05$.

[OBS] Panda still wins on the hetero_controlled subset (+0.261 at H = 96, +0.309 at H = 336; both significant). The advantage is smaller than on the homo_matched subset (+0.369 and +0.305 respectively) but remains statistically significant. [OBS] Comparing hetero_controlled against homo_matched from Experiment 25: at H = 96, Panda MAE is 0.605 (hetero) versus 0.331 (homo). Panda MAE nearly doubles despite matched difficulty CV. [OBS] Chronos MAE increases proportionally from homo to hetero (0.700 → 0.866), while Panda MAE increases more steeply (0.331 → 0.605). Panda degrades approximately 1.8× more than Chronos.

[PAT] After controlling for both mean difficulty and difficulty variance, Panda absolute MAE is substantially higher on heterogeneous Weather channels than on homogeneous channels at matched difficulty. Chronos does not show this differential sensitivity. This is the most controlled evidence that sensor heterogeneity is a specific bottleneck for Panda.

6.3.4 Competing Explanations

1. [HYP] Panda's joint channel attention actively harms performance on heterogeneous channels by averaging incompatible channel representations. The channel attention was pretrained on homogeneous ODE state variables and imports coupling patterns that are not present in heterogeneous sensor data. **Confidence: Medium.**.
2. [HYP] The difficulty matching controls are incomplete. Some third property of heterogeneous channels (multi-scale frequency content, non-stationarity patterns, distributional shift) is not captured by Chronos univariate MAE or its CV, and this third property is the true cause of Panda degradation. **Confidence: Medium.**.

Confidence: Medium. for the heterogeneity bottleneck pattern. The three-experiment heterogeneity series (24, 25, 26) provides consistent convergent evidence. **Confidence: Low.** for the proposed mechanism (joint attention importing incorrect coupling).

6.4 Chronos Heterogeneity Calibration Experiment

6.4.1 Motivation

To separately characterise how Chronos MAE behaves across heterogeneity levels, to confirm the interpretation that heterogeneity differentially affects Panda and not Chronos.

6.4.2 Method

Chronos run independently on the three difficulty-matched subsets from Experiment 25. Chronos forecasts each channel univariately and has no cross-channel attention, so it cannot be directly harmed by channel heterogeneity per se.

6.4.3 Observations

Subset	Het.	H	Chronos MAE	Difficulty
homo_matched	0.086	96	0.776	0.800
homo_matched	0.086	336	1.077	0.800
mixed_matched	0.602	96	0.817	0.839
mixed_matched	0.602	336	0.867	0.839
heterogeneous	0.947	96	0.687	0.839
heterogeneous	0.947	336	1.089	0.839

Table 20: Chronos heterogeneity calibration. Chronos MAE is approximately stable across heterogeneity levels at matched difficulty, as expected for a univariate model.

[OBS] Chronos MAE is approximately stable across heterogeneity levels at matched difficulty (0.776, 0.817, 0.687 at $H = 96$ across homo, mixed, heterogeneous). No monotone trend. [OBS] As expected, Chronos MAE scales with difficulty rather than heterogeneity, since Chronos has no cross-channel mechanism. [OBS] The contrast with Panda from

Experiment 25 is direct: Panda MAE increases monotonically (0.331, 0.475, 0.618 at H = 96) while Chronos MAE does not $(0.776, 0.817, 0.687)$. The divergence is attributable to Panda's multivariate architecture responding to heterogeneity.

Confidence: High. for the observation that Chronos is insensitive to heterogeneity. **Confidence: Medium.** for the implication that Panda's sensitivity is architectural.

6.5 Experiment 27: Burgers Univariate Ablation

6.5.1 Motivation

Experiment 10 showed Panda wins on Burgers at non-chaotic viscosity values ($\nu = 1.0$ and $\nu = 0.5$). Two competing explanations were: (1) Panda's channel attention captures spatial coupling in the PCA modes that is present even in non-chaotic flows, or (2) Panda's temporal architecture (Koopman lifting, non-causal encoder) is sufficient and channel attention is irrelevant. This experiment directly tests whether removing cross-channel attention (univariate Panda) changes the Burgers advantage.

6.5.2 Method

Multivariate Panda (16 PCA channels jointly) versus univariate Panda (each PCA channel independently). Tested at $\nu \in \{2.0, 1.0, 0.5\}$, covering the non-chaotic and transition regimes. $n_{\text{windows}} = 8$. $H = 96$.

6.5.3 Observations

ν	Panda multi	Panda uni	Adv multi	Adv uni	Δ	p (multi / uni)
2.0	0.0094	0.0078	+0.006	+0.001	−0.002	0.055 / 0.273
1.0	0.0165	0.0126	−0.002	+0.003	−0.004	0.320 / 0.004*
0.5	0.0235	0.0241	+0.033	+0.023	+0.001	0.004* / 0.004*

Table 21: Experiment 27. Adv = Chronos MAE − Panda MAE. Δ = Panda uni − Panda multi. * denotes $p = 0.004$, minimum achievable with $n = 8$ windows.

[OBS] At $\nu = 2.0$: multivariate Panda advantage (+0.006) is not significant (p = 0.055). Univariate Panda advantage (+0.001) is also not significant (p = 0.273). Both conditions are near zero; no conclusion is possible. **[OBS]** At $\nu = 1.0$: multivariate Panda advantage is −0.002 (Chronos wins marginally, p = 0.320, not significant). Univariate Panda advantage is +0.003 (p = 0.004, significant). Univariate Panda is statistically significant at this viscosity; multivariate Panda is not. **[OBS]** At $\nu = 0.5$: both multivariate (+0.033, p = 0.004) and univariate (+0.023, p = 0.004) Panda win significantly. The difference between multi and uni ($\Delta = +0.001$) is negligible. **[OBS]** Univariate Panda has lower absolute MAE than multivariate Panda at $\nu = 2.0$ and $\nu = 1.0$ (0.0078 vs 0.0094; 0.0126 vs 0.0165). Multivariate Panda has very slightly lower MAE at $\nu = 0.5$ (0.0235 vs 0.0241).

6.5.4 Critical Finding

[PAT] Channel attention does not drive Panda's advantage on Burgers at non-chaotic viscosities. At $\nu = 1.0$, univariate Panda outperforms multivariate Panda (lower absolute MAE and significant advantage versus Chronos, while multivariate is not significant).

At $\nu = 0.5$, both are equivalent. This is structurally parallel to the Weather univariate ablation result (Experiment 9), which also found univariate Panda competitive with or marginally better than multivariate Panda.

6.5.5 Competing Explanations

1. **[HYP]** The temporal architecture of Panda (Koopman patch embedding and non-causal encoder) is the primary driver of the advantage on both Burgers and Weather. Channel attention is adding noise rather than signal on these datasets. **Confidence: Medium.**: consistent across both ablations. Mechanism not isolated.
2. **[HYP]** Multivariate Panda underperforms univariate Panda at low-viscosity Burgers because the PCA modal time series are globally orthogonal (by construction of PCA). Panda's channel attention may try to learn coupling between orthogonal modes, which produces incorrect representations. **Confidence: Medium.**: specific to the PCA observable; not generalisable to raw spatial locations.
3. **[HYP]** With $n = 8$ windows, the multivariate versus univariate comparison at $\nu = 1.0$ (difference of 0.004 MAE units) is below the resolution of the test. The true advantage of multivariate Panda could be small and positive, undetectable at this sample size. **Confidence: Low.**: the direction at $\nu = 1.0$ consistently favours univariate across both MAE and advantage metrics.

Confidence: Medium. for the observation. Channel attention is consistently non-contributory or mildly detrimental on both Burgers and Weather, across two independent ablation experiments with different data characteristics. **Confidence: Low.** for the temporal architecture hypothesis, since no ablation has directly tested the temporal component.

7 Summary of Findings

7.1 Solid Observations (High Confidence)

1. Panda has statistically significant MAE advantage over Chronos on Weather at H = 96, 192, 336 ($n = 20$, $p \leq 0.001$). Advantage magnitude 0.17–0.24.
2. Channel attention does not drive Panda's advantage. Univariate Panda performs comparably to or marginally better than multivariate Panda on Weather (Experiment 9) and on Burgers at non-chaotic viscosities (Experiment 27). Both ablations show univariate Panda with lower absolute MAE than multivariate in the non-chaotic regime.
3. Panda wins on all tested chaotic `dysts` and Lorenz systems. Panda also wins on the Harmonic oscillator (+0.370, p = 0.004), directly contradicting the hypothesis that chaos is necessary for Panda's advantage.
4. Prediction head fine-tuning at H = 96 worsens Panda MAE by 0.091, ruling out the prediction head as a lightweight recoverable bottleneck.
5. Permutation entropy at order 3 cannot discriminate periodic from chaotic dynamics in any tested context (Lorenz sweep, complexity continuum, or cross-dataset).
6. After controlling for both mean difficulty and difficulty variance (CV), Panda MAE increases monotonically with sensor heterogeneity while Chronos MAE does not. This convergent evidence across three experiments (24, 25, 26) constitutes the strongest mechanistic signal identified in this investigation.

7.2 Medium Confidence Observations

1. Both Panda and Chronos degrade approximately equally when the periodic component is removed from Weather (Experiment 20). The earlier interpretation that Panda specifically depends on the full mixed signal is revised: both models depend on it, and projection error likely explains the collapse of advantage in Experiments 11 and 17.
2. Panda wins on Burgers at non-chaotic viscosities ($\nu = 1.0$, $\nu = 0.5$), and this advantage is driven by the temporal architecture rather than channel attention.
3. The complexity continuum shows a non-monotone advantage pattern. The Van der Pol limit cycle has anomalously small advantage (+0.011); the Harmonic oscillator has the second-largest advantage (+0.370). The chaos-specific hypothesis is insufficient.
4. Panda's advantage on Lorenz $\rho = 28$ partially collapses on a phase-shuffled surrogate with matched power spectrum, but this result is underpowered ($n = 8$).
5. Diversity subsampling advantage over stratified subsampling is not reproducible across seeds (Experiment 16). Panda absolute MAE is invariant to subsampling method; Chronos drives the observed differences.

7.3 Not Established Despite Earlier Discussion

1. The positive mechanism behind Panda's Weather and Burgers advantage. Channel attention is ruled out; the prediction head is ruled out as a recoverable bottleneck. The Koopman lifting and patching remain untested. These require code-level access to ablate.
2. Whether chaos is a sufficient condition for Panda's advantage. Panda wins on the

Harmonic oscillator and on Burgers at non-chaotic viscosities. The chaos-specific hypothesis is partially contradicted.

3. Whether the sensor heterogeneity bottleneck is due to channel identity (missing node embeddings) or interaction structure (missing graph topology). The node embedding proxy experiment (Experiment 22) shows only directional, small effects insufficient to distinguish these.
4. Whether any modification to Panda's architecture would improve generalisation. All component experiments identified bottlenecks or ruled out candidate mechanisms; no proposed intervention has been shown to recover performance.
5. Whether the Spearman correlation between λ_1 and Panda advantage is real (Experiment 4 p-hacking issue).
6. Whether λ_1 can be reliably estimated from Burgers PCA components (Experiment 15 estimator failure).

8 Proposed Next Experiments and Architectural Directions

8.1 Open Question: Koopman Lifting Ablation

This is the highest-priority unresolved question. The temporal architecture (Koopman random feature lifting, Takens-motivated patching, non-causal encoder) is the remaining candidate mechanism for Panda's advantage. No experiment has isolated it because ablating it requires modifying Panda's source code—replacing the random Fourier and polynomial feature lifting with a linear projection of equal dimension and comparing performance.

Proposed experiment: With code access, replace the Koopman lifting with a learned linear projection of the same output dimension. Evaluate on Weather, ETTh1, and Burgers at representative horizons. If the advantage collapses, the lifting is the positive mechanism. If the advantage is unchanged, the advantage resides in patching or the encoder attention itself.

What this would establish: The first positive causal evidence for an architectural component rather than a correlational observation.

8.2 G-SWaN-Style Node Embeddings (Primary Motivated Direction)

The convergent heterogeneity evidence (Experiments 24, 25, 26) identifies sensor heterogeneity as a specific bottleneck for Panda not shared by Chronos. Prabowo et al. (G-SWaN) propose per-sensor node embeddings combined with a spatial graph transformer for adaptive pairwise attention, motivated by exactly the property that real sensors are not interchangeable.

The scalar offset experiment (Experiment 22) is an insufficient proxy for node embeddings: it encodes only a scalar per-channel constant rather than a learned representation vector. A proper node embedding would be a learned vector $\mathbf{e}_i \in \mathbb{R}^d$ per sensor, injected into the channel attention computation.

Proposed experiment: Test whether adding per-sensor learned embeddings to Panda's channel attention (requiring code access) reduces the heterogeneity gap identified in Experiments 24–26. If Panda's relative disadvantage on heterogeneous versus homogeneous subsets decreases after adding node embeddings, the sensor identity hypothesis is supported. If the gap is unchanged, graph topology (not identity) is the missing component.

What would falsify the direction: If the heterogeneous–homogeneous Panda MAE gap is unchanged after adding proper node embeddings, the bottleneck is interaction structure rather than identity. In that case the XXLTraffic evaluation and a full graph-attention modification become the next motivated step.

8.3 Evaluation on XXLTraffic

XXLTraffic is a spatiotemporal traffic dataset with measurable sensor heterogeneity at a large scale (many sensors with known geographic and functional relationships). Evaluating Panda and a G-SWaN baseline on XXLTraffic would provide an out-of-distribution test of whether the heterogeneity bottleneck identified on Weather generalises to a structurally different real-world domain.

8.4 Persistent Homology Topology Analysis

The correlation dimension estimates from the topology analysis (Experiment 21 auxiliary) show a suggestive pattern: Weather and Lorenz cluster at $d \approx 0.86$–0.89, while ETTh1 and ETTh2 are at $d \approx 1.54$–1.62. However, the correlation dimension estimator is known to be unreliable on multivariate real-world data with limited observations.

Persistent homology on Vietoris-Rips filtrations of the state-space trajectories would provide a more rigorous topological characterisation. If Weather and Lorenz share topological features that ETTh does not, and if these features correlate with Panda's advantage, this would motivate the topological flow matching architectural direction suggested by Prof. Salim's group. The current correlation dimension result is suggestive but insufficiently reliable to motivate the direction without further analysis.